



University of Al-Ameed
College of Medicine



Internal medicine department

Psychology of Learning

الدكتور
علي لطيف السلطاني
شهادة البورد (الدكتوراه) في الطب النفسي
F.I.C.M.S(Psych.)

Learning Outcomes

1. Define learning and explain its significance in behavior change.
2. Identify and describe different types of learning, including non-associative and associative learning.
3. Understand the neural basis of learning and how it affects memory.
4. Apply knowledge of learning principles to clinical settings, particularly in relation to behavioral therapies.
5. Recognize the role of reinforcement and punishment in operant conditioning.
6. Differentiate between classical and operant conditioning, including their mechanisms and applications.
7. Discuss the importance of observational learning and its components.

- **Learning** is a relatively enduring or permanent change in behavior or knowledge that results from previous experience with certain stimuli and responses.
- The term behavior includes any observable response (fainting, salivating, vomiting).
- To understand why some learning is easy and some hard. We will see how psychologists identified three different principles that underlie three different kinds of learning: **classical conditioning, operant conditioning, and cognitive learning.**

Types of learning

1-Non-associative learning:

A relatively permanent change in the strength of response to a single stimulus due to repeated exposure to that stimulus.

Non-associative learning can be divided into habituation and sensitization

a. **Habituation:** is a gradual decrease in the response to a stimulus when it is repeated and is neither rewarding nor harmful.

- The organism learns: "This stimulus is harmless, so I don't need to respond strongly."

- **Habituation = Habit → less response**

- **Example**

A-A patient is admitted to the hospital and hears the **regular sound of the monitoring equipment**.

- First → the patient notices the sound.
- After hearing it repeatedly → the patient pays less attention to it.
- Eventually → the sound hardly bothers the patient.
- **Response decreases → Habituation**

B-A patient is admitted to the hospital and hears the **regular sound of the monitoring equipment**.

- First → the patient notices the sound.
- After hearing it repeatedly → the patient pays less attention to it.
- Eventually → the sound hardly bothers the patient.

- **B. Sensitization**: the progressive amplification of a response follows repeated administration of a stimulus.
- **Sensitization = Sensitive → more response**
- **Example**
 - A-A patient has a painful wound.
 - First → light touching causes some pain.
 - Repeated stimulation of the painful area → the patient becomes more sensitive.
 - Even a light touch may produce a stronger pain response.
 - B-A gentle touch initially causes mild discomfort.
 - Repeated touching of the sensitive skin → the person becomes more sensitive.
 - Even a light touch may produce a stronger pain response.

Response increases → Sensitization

2.Associative learning: This is the process by which an association between two stimuli or a behavior and a stimulus is learned.

- The two forms of classical and operant conditioning .
 - **Classical conditioning** is a type of learning in which a neutral stimulus gains the ability to produce a response that was originally elicited by another stimulus.
 - **Operant conditioning** refers to a kind of learning in which the consequences that follow some behavior increase or decrease the likelihood of that behavior's occurrence in the future.

3.Cognitive learning is a kind of learning that involves mental processes, such as attention and memory; may be learned through observation or imitation; and may not involve any external rewards or require the person to perform any observable behaviors.

A-Classical conditioning learning procedure in which two stimuli are paired – one (the conditioned stimulus) usually presented shortly before the other (the unconditioned stimulus) to produce a conditioned response to the first stimulus (learning).

- Pavlov noticed that his dogs began salivating before they received food.

- **Examples:**

Seeing the laboratory assistant who brought food.

Hearing the sound of the food bowl.

Recognizing the usual feeding time.

The dogs learned to associate these signals with food.

- **Neutral Stimulus**: Any stimulus that produces no conditioned response prior to learning.

- **Elements of Classical Conditioning** Pavlov identified several important elements required for classical conditioning. The terms unconditioned mean unlearned, while conditioned means learned.

- 1. Unconditioned Stimulus (UCS / US)** A stimulus that naturally produces an automatic and involuntary response without prior learning.

Example: Food naturally causes salivation in Pavlov's dogs.

- 2. Unconditioned Response (UCR / UR)** An automatic and involuntary response to an unconditioned stimulus. It is not learned and occurs naturally.

Example: Salivation in response to food.

- 3. Conditioned Stimulus (CS)** A previously neutral stimulus that becomes capable of producing a learned response after repeated pairing with an unconditioned stimulus.

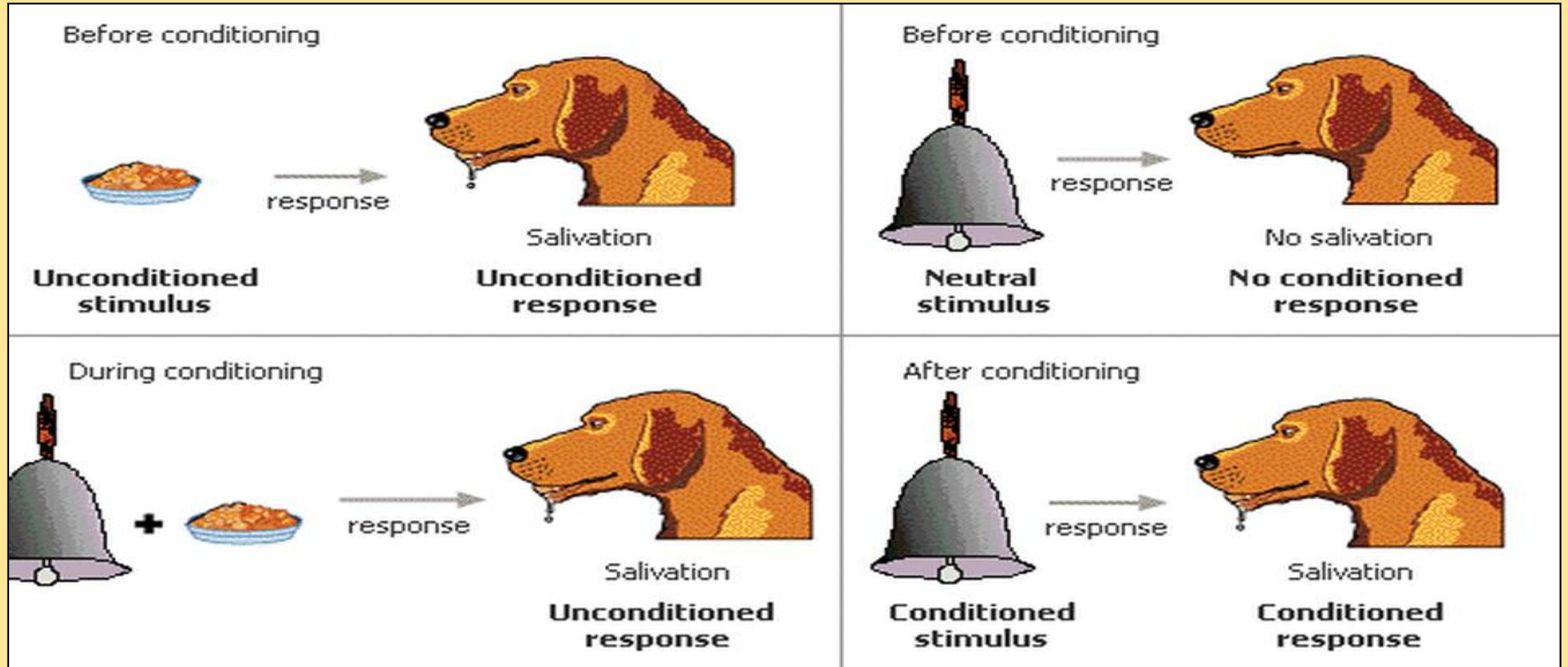
Example: Before conditioning: The food dish is a neutral stimulus and does not produce significant salivation. After repeated pairing with food: The dish produces salivation. The food dish is now a conditioned stimulus (CS).

- 4. Conditioned Response (CR)** A learned response produced by the conditioned stimulus. It is usually similar to the original unconditioned response but may be weaker.

Example: Salivation when the dog sees the food dish.

Pavlov's Experiment

Pavlov discovered that a neutral stimulus, when paired with a natural reflex-producing stimulus, will begin to produce a learned response, even when it is presented by itself.



- Sara experience in the dentist's office of being classically conditioned to feel anxious when she smelled a particular aftershave has some other interesting features.
- Sara found that similar odors also elicited anxious feelings but that other odors did not; her anxiety when smelling her boyfriend's aftershave gradually decreased, but accidentally meeting the dentist and smelling his aftershave triggered some anxiety.
- Pavlov found that these phenomena were associated with classical conditioning, and he named them generalization, discrimination, extinction, and spontaneous recovery.



2X DANDRUFF FIGHTING
POWER
BE UNBEATABLE

CLEAR **MEN**





- **Generalization** is the tendency for a stimulus that is similar to the original conditioned stimulus to elicit a response that is similar to the conditioned response.
- Usually, the more similar the new stimulus is to the original conditioned stimulus, the larger will be the conditioned response.
- Example: a new stimulus (e.g., a church bell) that resembles a conditioned stimulus (e.g., the lunch bell) causes a conditioned response (e.g., salivation).

- **Discrimination** occurs during classical conditioning when an organism learns to make a particular response to some stimuli but not to others.
- For example, Sara learned that the smell of a particular **aftershave** was associated with a painful dental procedure. Therefore, the smell of the aftershave caused her to feel anxious.
- However, the smell of her **nail polish** was very different from the aftershave and had never been associated with a painful dental procedure. Therefore, it did not cause anxiety.
- **Example:**
 - Aftershave smell → Anxiety
 - Nail-polish smell → No anxiety
 - Discrimination has **adaptive value** because it allows us to respond appropriately to different stimuli.
 - Generalization: Similar stimuli → similar responses.
 - Discrimination: Different stimuli → different responses.

- **Extinction** refers to a procedure in which a conditioned stimulus is repeatedly presented without the unconditioned stimulus and, as a result, the conditioned stimulus tends to no longer elicit the conditioned response.

Example

- Suppose a person has developed a fear of needles:
 - **Needle (CS) → Injection/pain (US) → Fear (CR)** During extinction, the person is repeatedly exposed to the **needle without the expected painful event**.
 - After repeated experiences:
 - **Needle → No painful event → Fear decreases**
 - Eventually, the needle may no longer produce a strong fear response.
- **Spontaneous recovery** is the tendency for the conditioned response to reappear after being extinguished even though there have been no further conditioning trials.
 - **Example: Aftershave → No painful procedure → Anxiety decreases → Extinction** .After some time, Ahmed smells the aftershave again, and her anxiety temporarily returns.

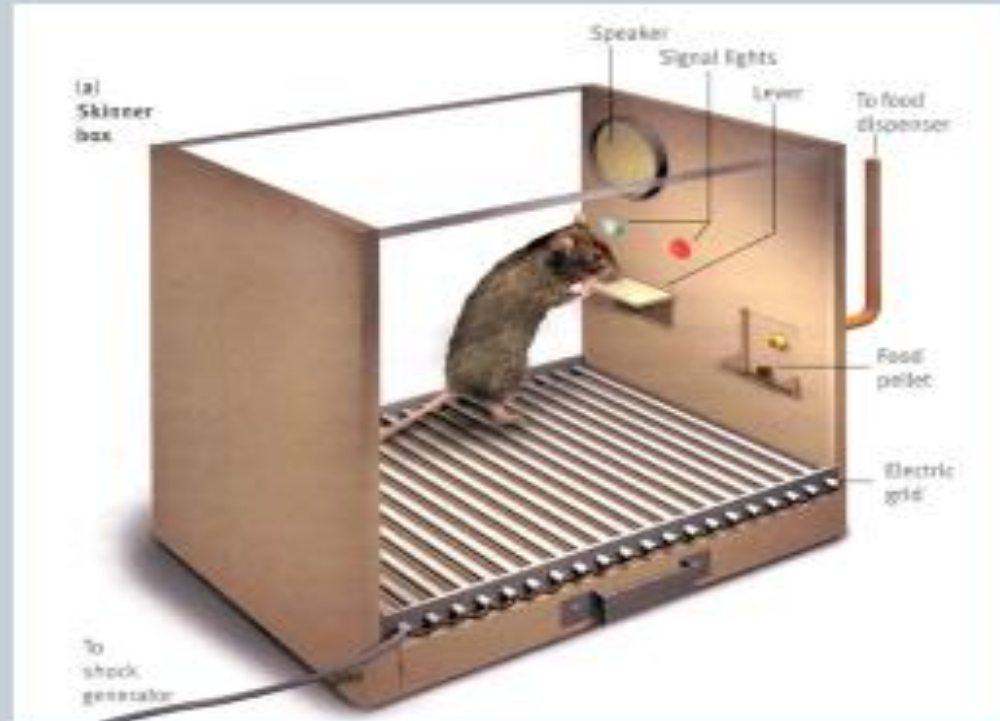
2. Operant Conditioning: B.F. Skinner

- **Operant conditioning**, also called instrumental conditioning, is a kind of learning in which an animal or human performs some behavior, and the following consequence (reward or punishment) increases or decreases the chance that an animal or human will again perform that same behavior.

B.F. Skinner

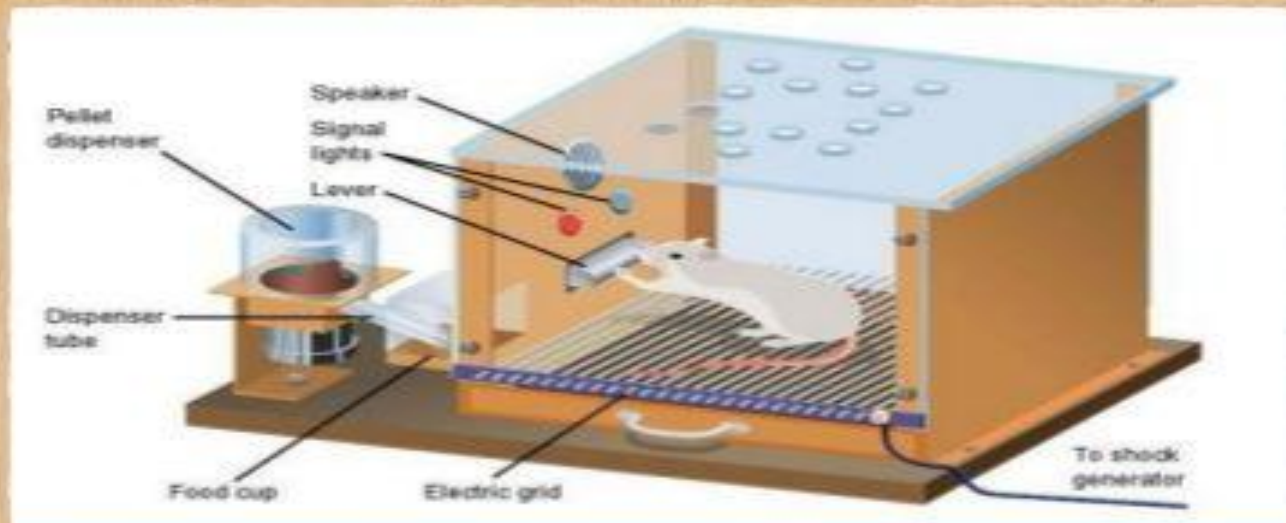


- B.F. Skinner became famous for his ideas in behaviorism and his work with rats.
- **Law of Effect:** The idea that responses that produced desirable results would be learned, or “stamped” into the organism.



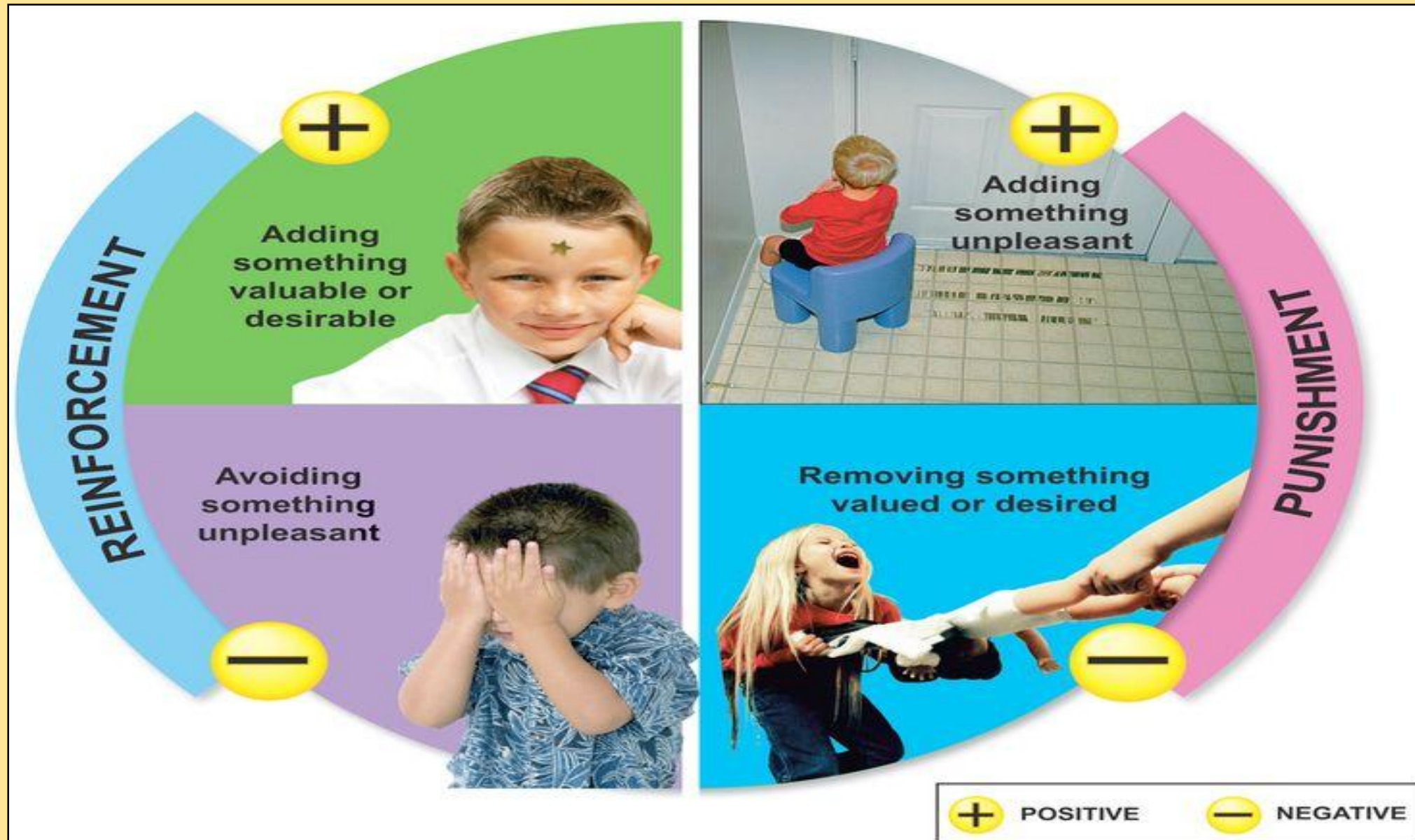
The Skinner Box

Skinner's operant conditioning chamber (also called a Skinner Box) was designed to teach rats how to push a lever. This behavior is not natural to rats, so operant conditioning with positive and negative reinforcement were performed in order to teach the behavior.



Positive Reinforcement:
A rat was awarded with food when he pressed the lever.

Negative Reinforcement:
A rat was able to turn off electric shocks produced by the floor by pressing the lever.



Operant Conditioning

Reinforcement

Increase behavior

Positive

Add appetitive stimulus
following correct behavior
Giving a treat when a dog sits

Negative

Escape
Remove noxious stimuli
following correct behavior
*Turning off an alarm clock by
pushing the snooze button*

Active Avoidance

Behavior avoids
noxious stimulus
Studying to avoid getting a bad grade

Punishment

Decrease behavior

Positive

Add noxious stimuli
following behavior
Spanking a child for cursing

Negative

Remove appetitive stimulus
following behavior
Sending a child to his room for cursing

1. Reinforcement is a consequence that occurs after a behavior and increases the chance that the behavior will occur again.

- There are two kinds of reinforcers .

A . Positive reinforcement : A behavior becomes more likely because it is followed by a pleasant consequence or reward.

- **Behavior → Pleasant stimulus added → Behavior increases**

- **Example:**

1-A student studies hard and gets a good grade.

- **Studying → Good grade → More studying**

2-Child with ADHD

- Child completes homework.
- Parents give **a reward or praise.**
- The child is more likely to complete homework again.

- There are two broad types of positive reinforcers:
- **Primary reinforcers** are stimuli, such as food and water, that an organism naturally finds reinforcing because they satisfy biological needs.

• شخص جائع - يأكل الطعام - يشعر بالشبع - يصبح الأكل أكثر احتمالاً عند الجوع.

- Food → satisfies hunger
- Water → satisfies thirst
- Sleep → satisfies need for rest
- Warmth → reduces cold



- **Secondary reinforcers** are stimuli that acquire reinforcing properties through their association with primary reinforcers.
- Money → associated with → food/other primary needs → becomes reinforcing



3-Example of Toilet Training

- Imagine you are the parent of **3-year-old Ahmed** who is developmentally ready to begin toilet training. Operant conditioning can be used to encourage the desired behavior.

1. Target Behavior Ahmed urinates in the toilet.

2. Preparation Before starting, reduce distractions by putting away her toys. Give her enough fluid, such as apple juice, so that she will need to urinate soon.

3. Reinforcement

- Choose a **reinforce**, such as: Candy or a small reward ,Verbal praise ,A hug
- Whenever Ahmed performs the desired behavior, immediately give her the reinforcer.

4. Shaping means reinforcing behaviors that gradually become closer to the final target behavior.

- For example:
- Ahmed says, "**I need to go to the toilet.**" → "That's great."
- He walks into the bathroom → "What a good boy".
- He lowers his pants by himself → "You're doing really well."
- He sits on the toilet → Praise or reward her.
- He successfully urinates in the toilet → Give strong praise, a hug, or a small reward.
- Thus, the child learns the complete behavior through **successive steps**.

- **B . Negative reinforcement** :A behavior becomes more likely because it removes or prevents something unpleasant.

- **Behavior → Unpleasant stimulus removed/avoided → Behavior increases**

Both *escape* and *avoidance* are the result of negative reinforcement.

1-Escape results when behavior terminates a negative reinforcer

- **Example:**

1-A person wears a seat belt to stop the annoying warning sound in the car. Removing the annoying sound reinforces the behavior of wearing the seat belt.

2-A patient with **panic disorder** experiences intense anxiety in a crowded shopping mall. The patient leaves the mall, and the anxiety quickly decreases.

- Because **leaving the mall removes the unpleasant anxiety**, the patient becomes more likely to leave the mall the next time anxiety occurs.
- **Anxiety → Leave the mall → Anxiety decreases → Leaving becomes more likely**

2-Avoidance means the unpleasant stimulus has not occurred yet, and the person performs a behavior to prevent or avoid it.

Example:

1-You know your phone battery will run out → you charge your phone before it reaches 0%.

You are more likely to charge it early next time.

2-A patient with social anxiety expects to attend a large meeting tomorrow.

Behavior: He tells his supervisor he is sick and does not attend.

Result: He avoids the anticipated anxiety.

Not attending is **negatively reinforced**, so he is more likely to avoid future meetings.

2. Punishment :Punishment is a consequence that occurs after a behavior and decreases the chance that the behavior will occur again.

- **There are two kinds of punishment or consequences—positive and negative—that decrease the occurrence of behaviors.**
- **A . Positive punishment** :refers to presenting an aversive (unpleasant) stimulus after a response. The aversive stimulus decreases the chances that the response will recur.
 - يُعطى طفل واجب إضافي نتيجة لسوء سلوكه في المدرسة. إضافة الواجب تعزز سلوك الالتزام بالقوانين
- **B . Negative punishment** :refers to removing a reinforcing stimulus (a child's allowance) after a response. This removal decreases the chances that the response will recur.
 - منع من استخدام الهاتف لمدة أسبوع بسبب عدم الامتثال للقواعد المنزلية. هذا يعزز احترام القواعد في المستقبل

Uses and Abuses of Punishment

- **Punishment rarely works in the long run for four reasons:**
 - Severe punishment may cause the child (or animal) to avoid the punisher instead of the behavior being punished, so the child (or animal) learns the wrong response.
 - Severe punishment may encourage lying to avoid the punishment again, not the response that is desired.
 - Severe punishment creates fear and anxiety, emotional responses that do not promote learning
 - Hitting provides a successful model for aggression

- The way to make punishment more effective involves remembering a few simple rules:
- 1. **Punishment should immediately follow the behavior it is meant to punish**
- 2. **Punishment should be consistent.**
 - There are two points:**
 - A. Keep your promise.**
If the parent says, “If you hit your brother, you will lose your toy,” the parent should actually do it.
 - B. Do not make punishment weaker randomly.**
 - For example:**
 - First time:** child jumps on the bed → scolding.
 - Second time:** child jumps on the bed → scolding or loss of a favorite toy.
 - If the punishment is sometimes severe and sometimes very mild, the child may **take a chance and continue the behavior.**
- 3. **Punishment of the wrong behavior should be paired, whenever possible, with reinforcement of the right behavior.**

2. Cognitive complex learning

- Cognitive learning, which involves mental processes such as attention and memory, says that learning can occur through observation or imitation and such learning may not involve any external rewards or require a person to perform any observable behaviors.
- **An organism's ability to mentally represent aspects of the world and then operates on these mental representations not on the world itself.**
- **Active form of learning**
- **No reward**

- It can occur in the following ways either :

A-Insight learning is a mental process marked by the sudden and unexpected solution to a problem: a phenomenon often called the

- “ah-ha!” experience. It is often described as an : “Aha!” moment

Everyday example:

- A person wants to reach a book on a high shelf but cannot reach it.
 - They look around and see a chair.
 - Suddenly they realize: **“I can stand on the chair!”**
 - They move the chair under the shelf and get the book.
- The solution appears **suddenly after understanding the relationship between the chair and the problem.**

Medical student example

- A medical student is given a patient with:
 - Excessive thirst
 - Frequent urination
 - Weight loss
 - High blood glucose
- At first, the student is unsure of the diagnosis.
- Then suddenly the student connects the findings:
- “Aha! This is diabetes mellitus.”

- **Trial-and-error learning:** The learner **does not know the correct solution at first.**
- They: Try → Fail → Try another response → Fail → Try again → Success ,After repeated attempts, the successful behavior becomes more likely to be used again.
- **Everyday example: Learning to use a new remote control**
 - A person wants to turn on the TV but does not know which button to press.
 - Press button A → nothing happens ✕
 - Press button B → changes the channel ✕
 - Press button C → TV turns on ✓
 - Next time, the person remembers and presses button C
- The person learned the correct response through **repeated attempts and mistakes.**

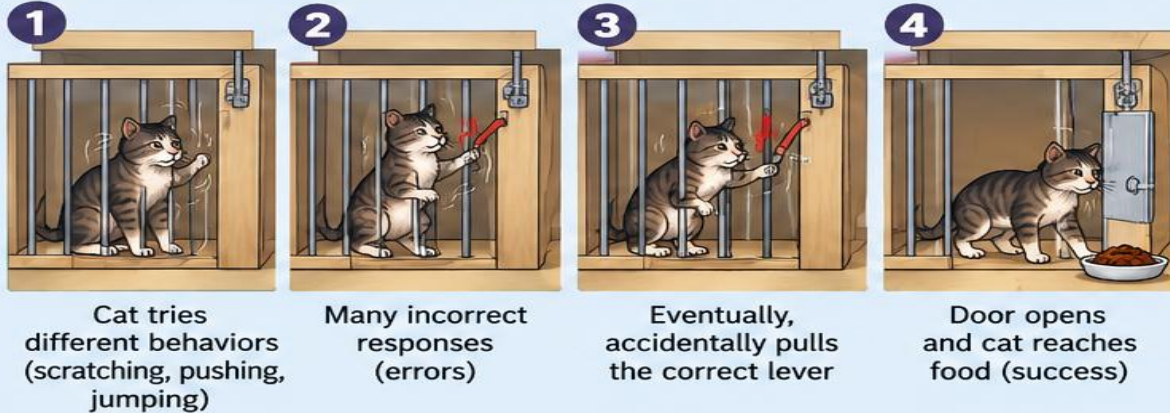
Medical example

- A medical student is learning to measure blood pressure manually:
 - First attempt → incorrect cuff placement ✕
 - Second attempt → incorrect positioning ✕
 - Third attempt → correct technique ✓
 - With practice → performs it correctly more quickly.
- Trial-and-error learning = gradually discovering the correct response through repeated attempts and errors.

Trial-and-Error Learning

Definition: Learning by trying different responses, making mistakes, and gradually learning which response produces the desired result.

Thorndike's Cat Experiment



With repetition, the cat escapes more quickly.

Key Features:

- Involves many attempts and mistakes
- Correct response is learned gradually
- Successful behavior is more likely to be repeated
- No sudden understanding

Insight Learning

Definition: A sudden understanding of the relationship between different parts of a problem that leads to a solution (often called an “aha!” moment).

Köhler's Box-and-Banana Experiment



Köhler's Stick Problem



Key Features:

- Sudden solution after understanding the problem
- Involves mental processes (thinking, perception of relationships)
- Not just repeated trial and error
- Often described as an “aha!” moment

3. Observational learning التعلم بالمحاكاة

- **Observational Learning** emphasizes the importance of observation, imitation, and self-reward in the development and learning of social skills, personal interactions, and many other behaviors.
- Unlike operant and classical conditioning, this theory says that it is not necessary to perform any observable behaviors or receive any external rewards to learn.
- Bandura believes that four processes—attention, memory, imitation, and motivation—operate during **Observational Learning**.

- For a person to learn by **observing another person (the model)**, four processes are important:

1. Attention (watch)

- The observer must **pay attention** to what the model is doing.
- **Example:**
A woman is afraid of spiders. She watches a person calmly holding a large spider.
(She must focus on the model's behavior)

2. Memory (Remember)

- The observer must **remember what they saw** so they can use it later.
- **Example:**
The woman remembers: (She stores this information in memory)

3. Imitation (Do)

- The observer must be able to **use the remembered information to perform the behavior**.
- **Example: Later**, the woman tries to hold a spider while remaining calm, just like the model.

4. Motivation (Want)

- The observer needs a **reason or incentive** to imitate the behavior.
- **Example:** The woman wants to overcome her fear because she wants to go **camping with her friends**.
- Therefore, she is motivated to practice holding spiders calmly.

1

Attention



The observer must pay attention to what the model says or does.

The woman (observer) watches a nonfrightened person (model) holding a large spider.

2

Memory



The observer must **store** or **remember** the information so that it can be retrieved and used later.

The woman will remember the image of the calm person holding the spider.

3

Imitation



The observer must be able to use the **remembered** information to guide his or her **own actions** and thus **imitate** the model's behavior.

The woman tries to imitate the model's calm behavior when holding a spider.

4

Motivation



The observer must have some reason or incentive to imitate the model's behavior.

The woman is motivated to overcome her fear of spiders because she wants to go on camping trips with her friends.

Watch**Remember****Do (Imitate)****Want (Motivation)**

Communication Skills

- This lecture supports the Communication Skills longitudinal theme by explaining how learning processes shape patient behavior, health beliefs, and responses to clinical advice.
- Understanding classical conditioning, operant conditioning, and observational learning helps students communicate in ways that reinforce positive health behaviors and modify maladaptive ones.
- For example, the lecture shows that reinforcement strengthens desired behaviors, punishment may be ineffective long term, and modeling and observational learning influence how individuals adopt behaviors from others .
- These principles guide clinicians in motivating patients, delivering health education, and supporting behavior change.
 - Applying this knowledge strengthens communication by enabling students to:
 - use encouragement and reinforcement when promoting treatment adherence
 - explain health behaviors in a way that supports learning and retention
 - model appropriate attitudes and communication styles for patients and families
 - recognize how past experiences and conditioning influence patient responses



Thank You

